

Cardiovascular disease identification using a hybrid CNN-LSTM model with explainable AI[☆]

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ABSTRACT

Cardiovascular disease (CVD) is a leading cause of death worldwide, with millions dying each year. The identification and early diagnosis of CVD are critical in preventing adverse health outcomes. Hence, this study proposes a hybrid deep learning (DL) model that combines a convolutional neural network (CNN) and long short-term memory (LSTM) to identify CVD from the clinical data. This study utilizes CNN to extract the relevant features from the input data and the LSTM network to process sequential data and capture dependencies and patterns over time. This study provides insights into the potential of a hybrid DL model combined with feature engineering and explainable AI to improve the accuracy and interpretability of CVD prediction. We evaluated our model on a publicly available dataset where the proposed CNN-LSTM achieved a high accuracy of 73.52% and 74.15% with and without feature engineering, respectively, in identifying individuals with CVD, which is the best result compared to the current state-of-the-art model. The results of this study demonstrate the potential of DL models for the early diagnosis of CVD. Our proposed CNN-LSTM model also incorporates explainable AI to identify the top features responsible for CVD. They could be used to develop more effective screening tools in clinical practice.

1. Introduction

Cardiovascular disease (CVD) is a condition that causes blood vessel obstruction and heart attacks with chest discomfort, as well as other heart diseases and heart failure that may result in death or other severe issues [1]. It is number one on the list of the top ten reasons for passing in the previous 15 years, and 15 million persons passed in 2015 [2]. January 2017 research demonstrated cardiovascular infections are the

main cause of death on a global scale. In 2020, according to the World Health Organization, this disease will be the primary cause of death worldwide, with an estimated 17.9 million fatalities annually [3]. Additionally, coronary disease mortality increases annually. It is anticipated that the population will surpass 23.6 million by 2030. CVD is considered the leading cause of death worldwide. Coronary heart disease, cerebrovascular disease, peripheral arterial disease, strokes and transient ischaemic attacks (TIA), vascular illness, chronic heart illness,

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congenital heart defects, thrombosis of deep veins, and pulmonary embolism are the most common cardiovascular conditions [4]. Several contributory risk factors, including hypertension, elevated blood pressure, excessive lipids, and abnormal pulse rate, complicate the diagnosis of cardiac disease. Detecting CVD at an early period is essential to reducing this toll.

Machine learning (ML) is one of the most swiftly evolving areas of Artificial Intelligence (AI). These algorithms can analyze vast amounts of data from numerous disciplines, including the vital medical field [5]. Moreover, data mining is among the numerous methods for enhancing disease detection and diagnosis. Furthermore, early detection of CVD reduces costs and CVD mortality. By employing an algorithm for classification, which contributes an essential part of medical research, data analysis strategies can efficiently and affordably complete the task. Using a variety of ML algorithms, such as Logistic Regression, Naive Bayes (NB), Support Vector Machines, Decision Trees, and others we constructed a model.

Through supervised learning and binary analysis of user data, we addressed the identified issue. By dividing the data into study and training sets, applying different methods or combinations, and evaluating for precision, we developed an accurate predictive model for assessing an individual's risk of CVD. Using this procedure, an adverse factor is identified with high accuracy. This study anticipated the dangers and etiology of CVD from one generation to the next. Informatics in healthcare for CVD is changing in various disciplines, including storing and sending data. The medical consequences of ML for CVD in 2020 are that it accesses the material in a supervised format and provides more accurate data for the anticipated condition.

1.1. Motivation

Late identification of the most critical behavioural risk factors for heart disease and stroke underscores the need for this study. Early detection of these conditions through routine screening exams could improve treatment options and longer survival times. Public health initiatives have been established to encourage communities to undergo regular screening for chronic conditions such as CVD. That is why we are approaching this study by using ML approaches. There are many algorithm-based studies, but we aim to conclude various algorithms in a point and express them with good accuracy. Here we employed the Support Vector Machine (SVM), Decision Tree (DT), K-Nearest Neighbors (KNN), NB, Multilayer Perceptron (MLP), CatBoost, Gradient Boosting Machine (GBM), AdaBoost, Random Forest (RF), and hybrid CNN-LSTM model. Based on our model and performance, we are motivated to conclude with the proposed CNN-LSTM method. Pre-screening systems for early disease prediction, detection, and prevention are a crucial source of information that applies cutting-edge ICT methods to address issue with health data collection, analysis, and interpretation, as well as to enhance current health systems for the advanced screening of diseases that we can explicitly discuss in this work. We aim to get over the previous restriction and produce a successful outcome.

1.2. Contribution

A summary of the primary contribution of our study includes the following:

- We introduce our novel CNN-LSTM model, achieving exceptional accuracy in early-stage cardiac condition diagnosis for patients. As deep learning pioneers, we are the first to develop such a model using CSV datasets.
- Furthermore, our model excels in both accuracy and efficiency, setting new standards. Utilizing CNNs and LSTM networks, we achieve faster diagnosis compared to current ML algorithms, facilitating prompt interventions and potentially improving patient outcomes.

- Additionally, we used feature engineering techniques to identify and incorporate relevant features such as 'blood_diff,' 'BMI,' 'obese,' and 'hypertension' into our model for CVD detection. Additionally, we employed preprocessing strategies to refine the dataset, enhancing its specificity and improving the overall performance of our model.
- Lastly, we utilize explainable AI techniques to identify the key features affecting cardiac patients in our CNN- LSTM model to gain valuable insights for clinical decision-making and improving patient outcomes.

1.3. Organization

The remaining sections of the paper are organized as follows: Section 2 provides a comprehensive literature review, offering an overview of existing research. The subsequent section 3, delves into the challenges associated with CVD detection. The methodology is outlined in Section 4, detailing the research design and data collection techniques. Sections 5 and 6 present the results and discussions, respectively, analyzing the findings and their implications. Finally, Section 7 concludes the paper, summarizing the main conclusions and proposing directions for future research.

2. Literature review

Cardiovascular identification using ML techniques is expanding rapidly, with numerous studies investigating the application of ML algorithms and DL models. By utilizing these techniques, healthcare can be revolutionized by enabling more precise diagnoses and tailored treatment plans for patients. This section reviews prior studies on identifying CVD through diverse ML approaches.

In [6], the authors proposed an RF algorithm correlating diabetes and heart disease. The method employed to determine the percentage of heart disease prediction is aware of the connection and the extent to which diabetes affects coronary artery disease. Better performance can be obtained by using more parameters. The authors in this study [1] concentrated on leveraging healthcare data for cardiovascular disease prediction through a mobile-based iOS application, obtaining an impressive accuracy of 72.7%. They propose extending the model to encompass other diseases and exploring deep learning and CNN methods for potential efficiency enhancements.

The authors [7] proposed a model employing four methods of classification for data mining: KNN, NB, DT, and RF. They utilized numerous data mining techniques, including regression, clustering, and association rules, to retrieve valuable data and details from large databases. They discovered the maximum accuracy with KNN ($k = 7$). Implementing more complex models and incorporating additional data mining techniques, including time-series analysis, the combination of association and clustering rules, SVM, and genetic algorithms, can improve the accuracy of early heart disease prediction. In Ref. [8], the authors applied several ML techniques, namely, DT, NB, and Neural Networks, and developed a prototype of the Intelligent Heart Disease Prediction System (IHDPSS). This approach has improved accuracy and also saved time and money. They observed that the accuracy of NB was 60%, logistic regression was 61.45%, and SVM was 64.4%. The authors [9] employed a method to boost the coronary artery disease prediction rate more correctly. They utilized effective prediction techniques, such as Gaussian NB, SVM, RF, Hoeffding Tree, and LMT, to boost accuracy. RF gives the most accurate results among all the algorithms used. Adding more input attributes and reducing the data size can enhance the genetic algorithm's performance. In Ref. [10], the authors proposed a method for predicting cardiac disease utilizing sklearn libraries, pandas, matplotlib, and other required libraries with a TkInter Python-based application. The hybrid model achieved 88% accuracy. Applying DL techniques to predict heart disease may yield improved results. The authors [11] analyzed heart disease prediction via an ML technique. Comparing all the algorithms used in this method, the RF shows the best

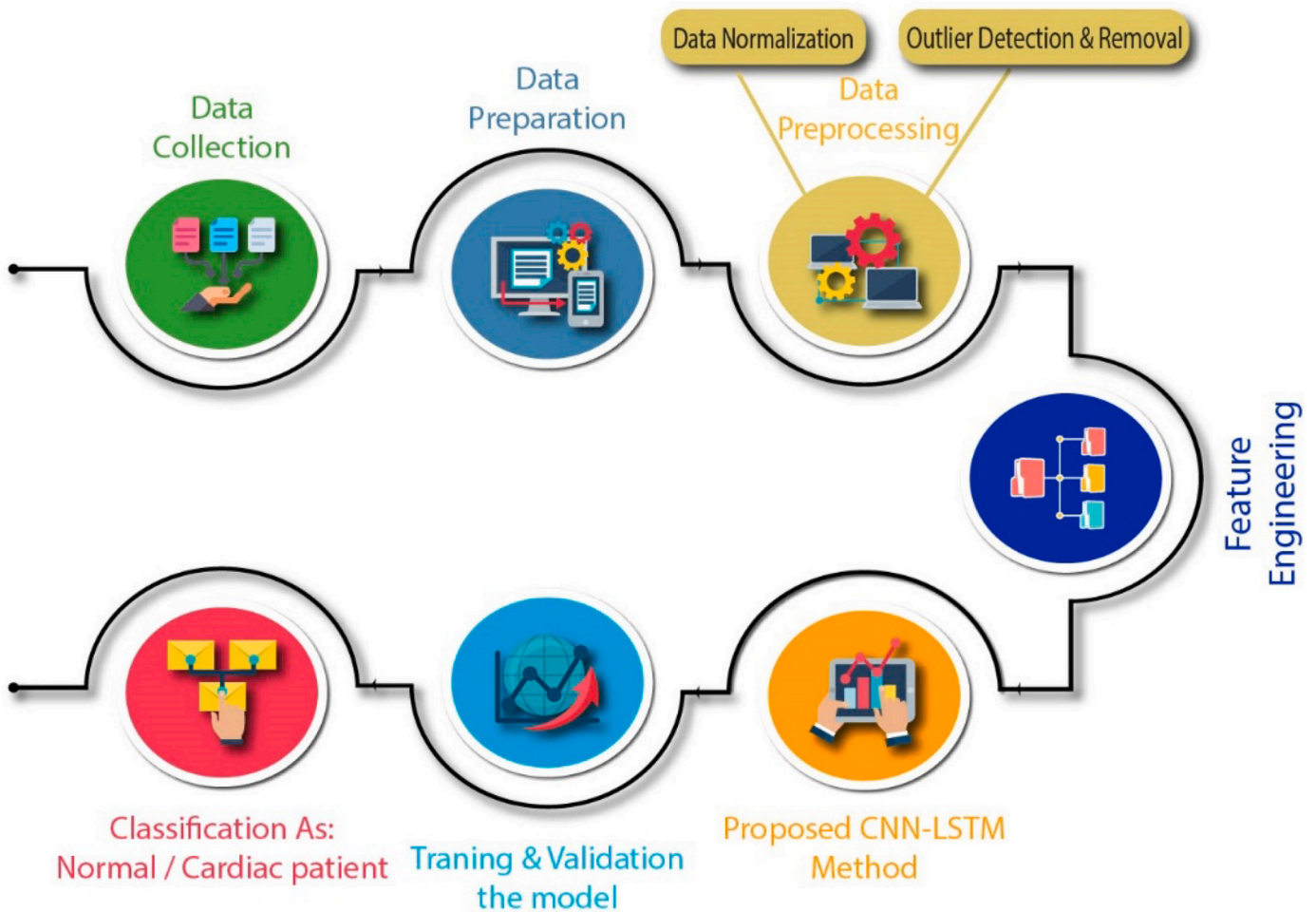


Fig. 1. Workflow of our proposed methodology.

result of 95.60%. The results can be improved by implementing other algorithms and fruitful deep-learning techniques. In Ref. [12], the authors proposed a method for evaluating and summarizing the overall predictive capability of ML algorithms in CVD. They demonstrated that the predictive ability of ML algorithms in CVD, particularly SVM and boosting algorithms, is promising. They might obtain an optimal accuracy rate using more feature selection methodologies and techniques. The authors [13] employed a hybrid approach using ML classifiers for CVD forecasts. They have constructed their framework using a neural network based on general principles. Using the confusion matrix, the proposed method employing the ML classifier obtained a higher accuracy of 85.71%. By varying the number of testing datasets, accuracy can be increased.

In [14], sonam et al. developed a heart disease prediction system using two classification techniques: NB and DT classifiers. The DT classifier outperformed the NB classifier, but removing irrelevant attributes from the dataset improved the NB classifier's performance. This research emphasizes the importance of classification techniques and data preprocessing for accurate heart disease prediction. In Ref. [4], the authors proposed a method for predicting CVD using symptom input. Six classification algorithms were used to analyze 14 attributes of the Cleveland dataset, with SVM and MLP achieving the highest accuracy of 91.7%. Ensembles and exploring more parameter settings could further improve performance. This method shows promise in providing accurate and immediate disease prediction. The authors [15] employed a Convolutional Neural Network (CNN) approach for predicting CVD. Traditional ML methods, such as Logistic Regression, K-Nearest Neighbors (KNN), NB, Support Vector Machine (SVM), and Neural Networks (NN),

are compared to the proposed CNN model. The paper claims a high accuracy of 94% for the CNN model using the UCI ML repository dataset. The CNN model aims to improve efficiency and accuracy by leveraging DL capabilities to analyze complex patterns in medical data. In Ref. [16], the authors addressed the global health burden of CVD, specifically coronary heart disease. The authors proposed an enhanced deep neural network (DNN) model for accurate and reliable diagnosis. The model achieved 83.67% diagnostic accuracy, 93.51% sensitivity, 72.86% specificity, and 79.12% precision. Investigating enhanced DL methods and advanced models can be effective in further improving the accuracy of heart disease diagnoses in patients worldwide. In Ref. [17], the author of the paper employed a model using DL Techniques (DLTs), specifically Artificial Neural Networks (ANN), to analyze and predict CVD in the Robust Healthcare Industry (RHI). The ANN model achieves a high accuracy of 98.4% compared to other models, highlighting its effectiveness in disease analysis and prognosis.

The existing literature on CVD detection using ML algorithms reveals that several studies have been conducted in this domain. However, there is a research gap in developing more effective and productive models that can improve the accuracy and efficiency of these algorithms. Although most studies have focused on using ML algorithms for classification and prediction, the lack of research on developing efficient preprocessing techniques to enhance algorithm performance is evident. Although some studies have achieved high accuracy rates, there is still a need to improve the prediction of CVD at an early stage and identify more accurate risk indicators. There is no available related work to detect CVD using the DL method on tabular data. Furthermore, it has been observed that most researchers used a limited dataset, raising

Table 1
Details of dataset features.

SN	Attribute name	Description
1.	age	Age in days.
2.	gender	Sex indicate in 1(Woman); 2(Man);
3.	height	Denoted height in centimeters(cm)
4.	weight	Denoted weight in kilogram(kg)
5.	ap_hi	Systolic blood pressure at the time of heartbeats.
6.	ap_lo	Diastolic blood pressure at the time of heart rests between beats.
7.	cholesterol	Based on amount of lipoprotein variation in blood as 1: normal; 2: above normal; 3: well above normal;
8.	gluc	Fasting blood glucose level for patient in express as 1: normal; 2: above normal; 3: well above normal;
9.	smoke	whether the patient is a smoker (1) or nonsmoker (0).
10.	alco	Alcohol level in patient's blood that develops from drinking alcoholic beverages written as nonalcoholic(0); alcoholic (1);
11.	active	Physical activity either presence (1) or absent (0)
12.	cardio	Working capacity of hearts valves and chambers for targeting variable as no disease (0); disease (1);

concerns about their findings' generalizability. Additionally, the utilization of explainable artificial intelligence (XAI) and feature engineering techniques has not been reported in many studies, which can provide insightful information on the performance of a classifier.

Therefore, this study aims to address these gaps and enhance the development of effective and efficient ML models for CVD prediction.

3. Challenges of CVD detection

Though this is a successful appearance of the model, we are facing some challenges as well. New charts for 21 geographical regions have been produced by the WHO CVD Risk Chart Working Group study to aid in risk prediction in clinics and national public health initiatives [18]. CVD is a significant cause of death and disability worldwide, and early detection and treatment are crucial for preventing its progression. ML algorithms have shown promise in detecting CVD, but several challenges must be addressed to improve their accuracy and reliability. The development of population-specific models that fully address the issues of present risk prediction models will require further resources and efforts to collect more observational data with extensive follow-ups. Risk prediction methods are undoubtedly at the height of development.

- **Lack of standardized data:** ML algorithms require large amounts of high-quality data for training and validation, but there is a lack of standardized data in CVD detection. The data may need to be completed, accurate, or consistent, which can affect the performance of ML algorithms.
- **Imbalanced data:** The distribution of CVD and non-CVD cases in the training data may be imbalanced, with most claims being non-CVD. This can lead to biased results and poor performance of ML algorithms in detecting CVD [19].
- **The complexity of CVD:** CVD is a complex disease with many risk factors and symptoms, which makes it challenging to identify the most critical features for detection. ML algorithms may also be challenged by commodities and confounding factors that can affect the accuracy of the results.
- **Ethical considerations:** Using ML algorithms in healthcare raises ethical concerns like privacy, informed consent, and bias. The algorithms should be ensured that they are transparent, unbiased, and do not violate ethical principles essential.
- **Interpretability:** In ML algorithms, black boxes are often considered, making it difficult to interpret the results and understand how the algorithm arrived at a particular decision. This makes CVD detection difficult, where the underlying factors and mechanisms still need to be fully understood.

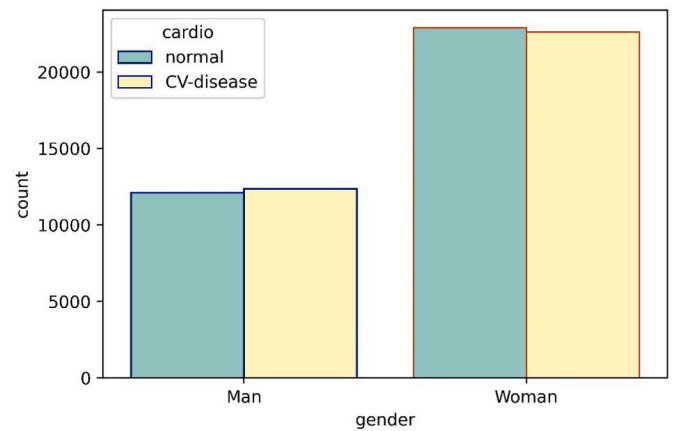


Fig. 2. The number of diseased and non-diseased people.

4. Materials and methodology

We divide this system model of methodology into several stages: data collection, data normalization, feature engineering, model architecture design, model training, hyperparameter tuning, and model evaluation, as shown in Fig. 1.

4.1. Materials

4.1.1. Dataset

In this work, a dataset on heart disease was analyzed to develop our anticipated model [20]. The dataset used in this study was obtained from the Kaggle CVD dataset. There are 14 attributes contained within this dataset. Table 1 illustrates the specifics of all features.

The data set comprises 70,000 patient records, comprising 24,470 males and 45,530 females of various ages, with 35,021 (50.03%) normal patients and 34,979 (49.97%) heart disease patients, as shown in Fig. 2. The descriptive statistics of the dataset are shown in Table 2.

4.2. Methodology

4.2.1. Data normalization

Normalization of data is a fundamental preprocessing stage for data mining and learning. It entails converting the data into an accepted format that can improve data precision and integrity, reduce duplication and inconsistency, and ensure that the data is arranged in an effective and consistent way [21]. Normalization is necessary when noticeable variations exist between the ranges of various features, and it is particularly appropriate when there are no outliers in the dataset. There are several data normalization techniques. The most commonly used normalization techniques are Decimal scaling, Unit Vector Normalization, Z-Score Normalization, Log Transformation, and Min-Max Normalization.

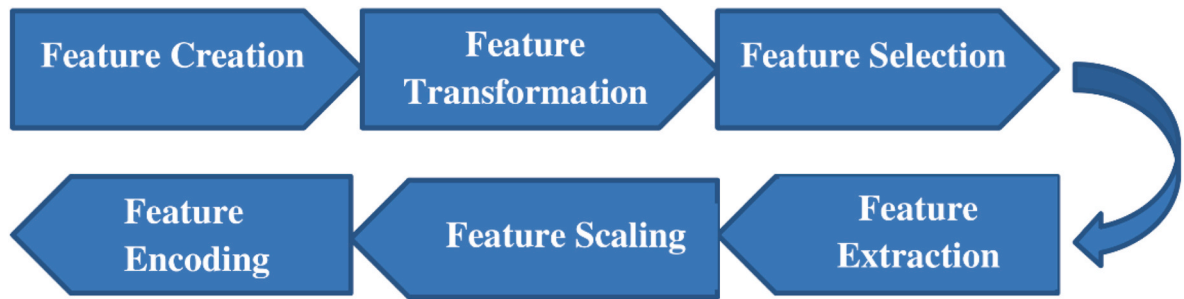
Our study used the min-max normalization technique to scale our data between 0 and 1. We are approaching this normalization technique because of its simplicity and effectiveness in enabling distance-measuring algorithms. It is essential to remember that the minimum and maximum values may not accurately represent the data in certain cases, potentially resulting in information loss. Below, we have provided a brief overview of the normalization techniques employed in our study.

Min-Max normalization balances value comparisons between data before and after the procedure by executing linear changes on the original data [22,23]. The data inputs used in min-max normalization are translated into a predetermined range (from 0 or 1 to 1). This standardization preserves the associations between the original data values. The method can be described in the following formula [24]:

Table 2

Descriptive statistics for the Dataset.

Variable	Min	Max	Mean	Median	STD	Variance	MAD	RMS	Skewness	Missing
age	30	65	53.338	54	6.765	45.769	5.644	53.766	-0.306	0
gender	1	2	1.349	1	0.477	0.227	0.455	1.431	0.631	0
height	55	250	164.359	165	8.210	67.406	6.324	164.564	-0.642	0
weight	10	200	74.206	72	14.396	207.237	11.044	75.589	1.012	0
ap_hi	-150	16020	128.817	120	154.011	23719.517	16.007	200.781	85.296	0
ap_lo	-70	11000	96.630	80	188.472	35521.894	31.044	211.799	32.114	0
cholesterol	1	3	1.367	1	0.680	0.462	0.549	1.526	1.587	0
gluc	1	3	1.226	1	0.572	0.327	0.385	1.653	2.397	0
smoke	0	1	0.088	0	0.283	0.080	0.161	0.297	2.906	0
alco	0	1	0.054	0	0.225	0.051	0.102	0.231	3.956	0
active	0	1	0.804	1	0.397	0.158	0.315	0.896	-1.529	0
cardio	0	1	0.500	0	0.500	0.250	0.500	0.707	0.001	0
blood_diff	0	1	44.899	40	10.658	113.589	8.282	46.147	0.833	0
BMI	3.472	298.667	27.464	26.298	5.609	31.471	4.032	28.031	5.581	0
obese	0	1	0.260	0	0.438	0.192	0.389	0.510	1.093	0
hypertense	0	1	0.814	0	0.389	0.151	0.303	0.902	-1.612	0

**Fig. 3.** Flowchart of feature engineering.

$$x_{new} = \frac{X - \min(X)}{\max(X) - \min(X)} \quad (1)$$

X = old value;

Max (X) = maximum value in the dataset;

The equation used is as follows:

$$x^* = \frac{x}{10_j} \quad (2)$$

where, $j = \log_{10}(\max(xi))$ (3).

4.2.2. Feature engineering

Feature engineering maximizes performance by extracting predictive data and choosing ML algorithms based on the feature sets [3]. It uses transform function, arithmetic, and aggregation operators to create new features and thus transforms a dataset's feature space to improve predictive modeling [25].

In this study, we have selected the most relevant features and created new features to improve the performance of our proposed CNN-LSTM model. Feature engineering includes the following steps shown in Fig. 3.

The workflow has a few steps:

1. Coefficient analysis can remove high-relevant descriptors.
2. Dimensionality enhancement using basic structural parameters and prototypical functions (such as multiply).
3. Top important descriptors selection to discover a good descriptor subset for material characterization [26].

ML requires laborious feature engineering, which helps DL and improved text classification rules learning. Our study incorporates a visual aid to illustrate the feature engineering process systematically.

Fig. 3 presents a flowchart outlining the steps in feature engineering. The flowchart provides a clear understanding of the process, starting with data collection, then data cleaning, feature selection, feature transformation, and finally, feature scaling. This approach helps to ensure that the most relevant and informative features are selected to improve model performance.

Moreover, we leveraged advanced feature engineering techniques to identify and incorporate crucial features like 'blood_diff', 'BMI', 'obese', and 'hypertension,' significantly enhancing the model's diagnostic capabilities. "Blood_diff" indicates the variability between systolic and diastolic blood pressures, while "hypertense status" determines hypertension. These features serve as crucial indicators for cardiovascular health analysis, identifying individuals at risk of hypertension-related cardiovascular conditions. Additionally, the standardized transformation of height and weight, known as "BMI," plays a significant role in assessing cardiovascular health and identifying potential obesity-related risks. By incorporating these features, our method enhances CVD detection accuracy and supports early intervention for better patient outcomes.

4.2.3. Feature creation

New features have been created using feature engineering techniques, including (blood_diff), Body Mass Index (BMI), obese, and hypertension. Creating new features by combining or interacting with existing variables can aid in capturing nonlinear interactions and enhance model precision.

$$\sum_{n_i \in T, f_{m=f}} \frac{E_{n_i}}{E} \text{InfoGain}(f_{n_i}, E_{n_i}) \quad (4)$$

When f is used as a splitting feature, its weighted information gain in that node is ascribed. When used as an argument of a splitting part, it is credited with the splitting feature's weighted info-gain, divided by the

total number of elements in its statements and discounted by. Demonstrates that the value of a developed feature should not be attributed only to its construction parts.

4.2.4. Feature transformation

Feature transformation involves converting or modifying raw input data into a more meaningful representation that can be used as input to an ML model and act as a better indicator.

In this paper, we have calculated (blood_diff) and hypertense status based on the (ap_hi) and (ap_lo) features, a feature transformation type in feature engineering. And BMI is a standardized and recognized transformation of features (height and weight). We determine obesity based on a person's BMI, a feature transformation in feature engineering. Several feature transformation methods exist, such as log transformation, Z-normalization transformation, scaled transformation, sine transformation, square-root transformation, etc. [27].

Log transformation translates feature values by applying the natural logarithm, a popular data transformation method. $y = \ln(1 + x)$, where y is the modified characteristic, x is the original characteristic, and $\ln(x)$ is the natural logarithmic function. Due to the limitation imposed by the $\ln(x)$ function, the above formula transforms only numbers greater than zero. A constant, such as the $\ln(1 + x)$ function, is introduced to address the zero values [28].

4.2.5. Feature selection

This is the process of picking a subset of the most relevant features from a more significant collection for use in a model. Employing too many features can lead to over-fitting and poor model performance. Feature selection is the process of producing a subset of an original feature set based on a set of feature selection criteria, which picks the pertinent features of a dataset. This study shows which feature is the most significant in different algorithms for predicting the disease more accurately.

4.2.6. Feature extraction

The feature extraction process transforms raw data into more relevant and useful characteristics for a particular application. Various techniques can be used for feature extraction, including convolutional methods that use manually designed kernels and syntactic and structural methods applied to sequential, geographic, or other structured data [29].

4.2.7. Feature scaling

This includes altering features to be of a comparable size, which might enhance the effectiveness of ML algorithms. Feature scaling is one of the most critical changes we must apply to our data [30]. Here in our paper blood_diff feature, we eliminate all the value which ranges above 80 because it is uncommon and which reduce the effectiveness of algorithms.

4.2.8. Feature encoding

The process of encoding categorical information into numerical form, using techniques such as one-hot encoding or ordinal encoding so that they may be utilized in an ML model [31].

4.3. Feature importance

In classical ML, the value of features is crucial in the data pre-processing [32]. "Feature importance" refers to methods that grade input features according to how well they predict a target variable. In predictive modeling, feature significance values are essential as they provide the basis for reducing dimensionality, insight into the data and model, and feature independence testing. A predictive model's effectiveness and efficiency can be increased through feature selection [32,33]. The feature (variable) significance indicates the impact of every feature in the model's prediction. It establishes the usefulness of a

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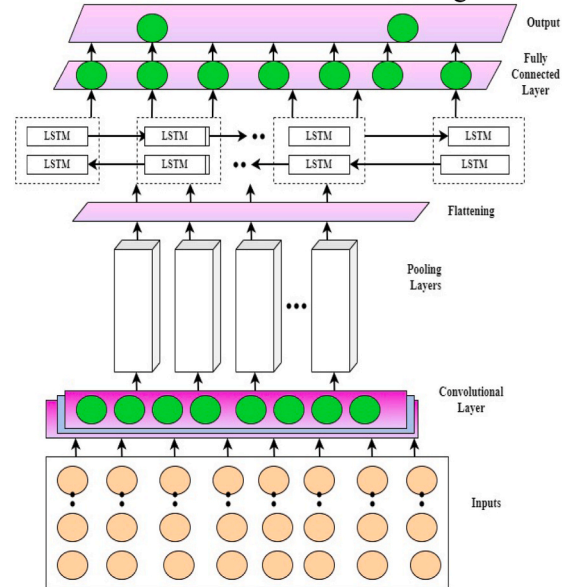


Fig. 4. Architecture of the proposed CNN-LSTM model.

particular variable for a given model and forecast. Although numerous methods for determining feature importance exist, they can generally be divided into model-independent and agnostic. Numerous model-neutral techniques have been created to increase ML models' openness, dependability, and interoperability. Interpreting feature relevance in supervised learning models can be difficult as attributes are interdependent. Such dependencies are disregarded by permutation feature importance (PFI), leading to incorrect extrapolation-based conclusions. Possibilities include more sophisticated conditional PFI techniques that permit the evaluation of feature relevance based on all other characteristics. It is advantageous if the conditioning is apparent and understood due to this change in perspective and permits accurate interpretations. The target outcome whether the cardiac disease is present's the same across datasets, despite variations in the numbers and types of characteristics. Not all information is equally essential for predicting heart disease. Some characteristics are more useful in predicting heart disease because of their relevance [3]. Using ML techniques, the significance of each feature was determined. According to their feature scores, the features were sorted. In our study of CVD identification using a hybrid CNN-LSTM model with explainable AI, feature importance analysis plays a crucial role. We performed individual feature importance analyses for each algorithm used in our study. Random Forest and Decision Tree algorithms provided feature scores, allowing us to determine the significance of each feature. However, KNN, SVM, and MLP algorithms did not generate feature scores [34], so we relied on other techniques to assess feature importance in those cases. Our proposed algorithm for CVD identification identifies ap_hi, ap_lo, and age as the top three features without feature engineering. However, with feature engineering, the top three features are age, ap_hi, and hypertense. These are the most significant features that represent the analysis result of the features by various algorithms.

4.4. Proposed CNN-LSTM model

The proposed CNN-LSTM model consists of two main components: a CNN for feature extraction and an LSTM for sequence modeling. The CNN component takes in the raw input data and extracts relevant features by applying convolutional filters. These filters capture local patterns in the data and can learn to identify important features [35]. The output of the CNN component is then fed into the LSTM component,

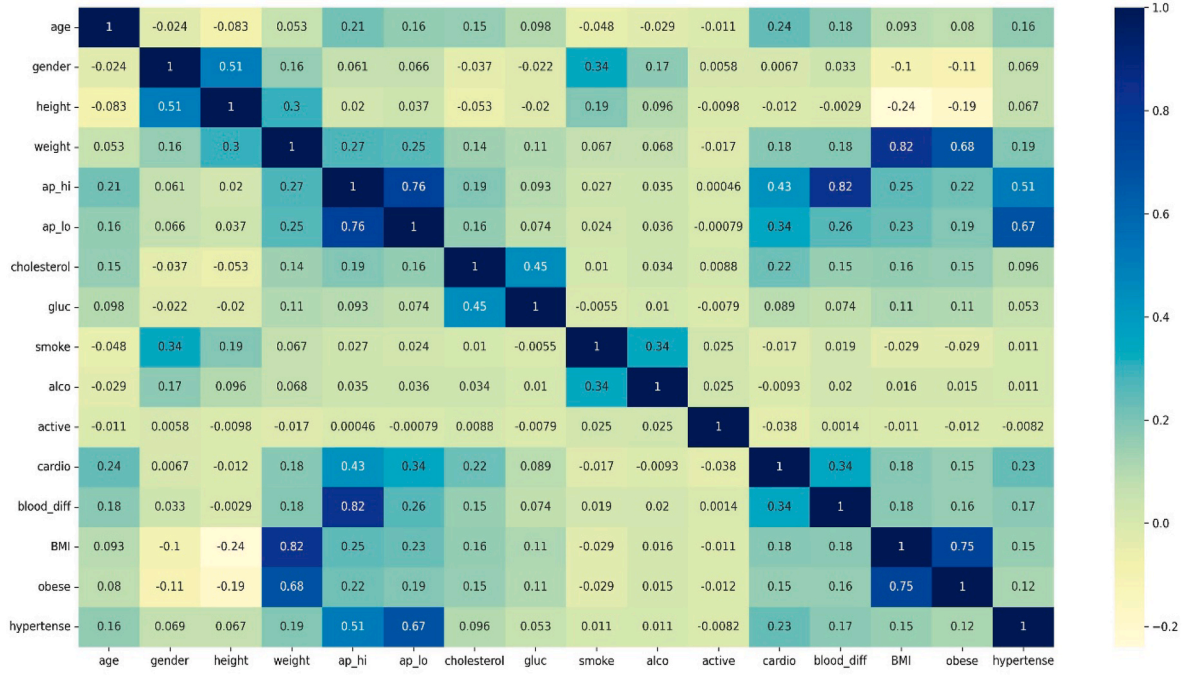


Fig. 5. Heatmap showing correlations among all the features of the dataset.

which processes the sequence of feature vectors to model temporal dependencies and capture long-term context [36]. The LSTM network contains memory cells that store information over time and gates that control the flow of information into and out of the cells. This allows the model to learn and remember patterns in the data that occur over long periods of time [37]. The final output of the model can be either a classification label or a prediction of the next value in the sequence, depending on the task at hand. The model is trained using back-propagation through time, which allows the gradients to flow through the entire sequence and update the model's parameters. The architecture of the model we proposed in this study is shown in Fig. 4.

The proposed CNN-LSTM model performs sequential data processing. Firstly, the CNN component extracts features automatically from the raw data, reducing the need for manual feature engineering. Secondly, the LSTM component can capture long-term dependencies and context, allowing the model to make more accurate predictions or classifications. Finally, the model can be trained end-to-end using back-propagation through time, making it more efficient and effective than traditional models that require separate feature extraction and sequence modeling steps.

In our study, we proposed a hybrid CNN-LSTM DL model for analyzing the dataset. Our proposed model introduces a hybrid CNN-LSTM DL model that leverages the strengths of both architectures. This allows us to capture spatial and temporal dependencies in the data effectively. Our approach fills a gap in the literature, as no previous work applies this hybrid model to the dataset. The integration of CNN and LSTM enables us to extract complex patterns and capture long-term dependencies, resulting in superior performance compared to existing methods.

In our proposed model for the CSV dataset, we adjust CNN parameters (number of layers, filter size, stride, pooling size), LSTM parameters (number of layers, units, dropout rate), training parameters (learning rate, batch size, number of epochs), feature engineering (feature selection, scaling), and model architecture (skip connections, attention mechanisms) can improve accuracy. Hyperparameters optimization methods like grid search or Bayesian optimization are essential for finding optimal combinations. Thorough experimentation and validation are crucial for achieving the best performance.

4.5. SHAP as XAI

SHAP (SHapley Additive exPlanations) is a well-known Explainable Artificial Intelligence (XAI) technique that describes ML model outcomes. It provides a method for measuring the impact of each model feature on the output of a specific data point. This can help to understand why a model made a particular prediction and which features were most influential in producing that prediction [38]. By utilizing SHAP, the insights of the model can be achieved into a model's predictions to allow for uncovering and rectifying biases, enhancing performance, and creating end-user confidence. Firstly, it calculates the SHAP values to see the insightful features that led to further steps [39]. The SHAP value can be determined via the formula below:

$$\varphi_i(f, a) = \sum_{z' \subseteq x} \frac{|z'|!(M - |z'| - 1)}{M} [f_x(z') - f_x(\frac{z'}{i})] \quad (5)$$

Where,

- φ_i = Shapley value for feature i
- f = Blackbox model
- x = input data point
- z' = subset
- x' = simplified data input
- $f_x(z')$ = With feature i .
- $f_x(\frac{z'}{i})$ = Without feature i .

It calculates the contribution of a single characteristic by considering all potential subsets of features that include that feature. The contribution is then weighted by the number of subgroups with the feature and averaged over all potential subsets. This produces a single SHAP value for each feature, which may be utilized to explain the model prediction for a specific instance.

Cardiovascular Disease Identification using CNN-LSTM

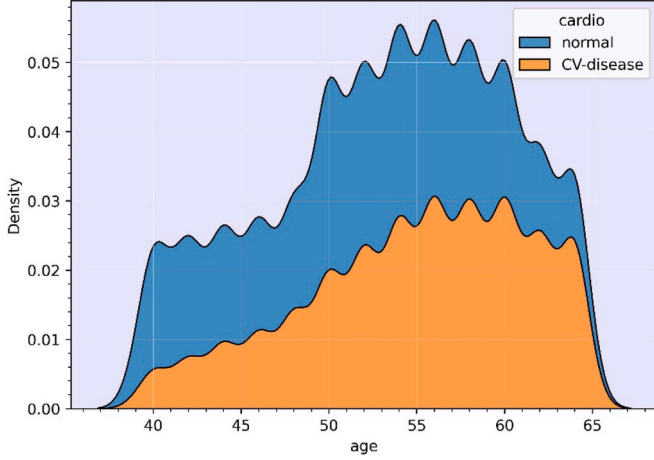


Fig. 6. KDE plot representing diseased and non-diseased patients based on the age distribution.

5. Results

5.1. Experiment setup

This study employed a conventional laptop with Windows 10, an Intel Core i7 processor, and 16 GB of RAM. The experiment was conducted five times, and the ultimate outcome was determined by taking an average of all five computational results.

5.2. Performance evaluation

The dataset underwent evaluation by eleven classification algorithms, which were then compared using a 5-fold cross-validation to determine the best approach based on accuracy and other statistical factors. The algorithms tested included GNB, SVM, DT, LR, CatBoost, AdaBoost, GBM, RF, MLP, XGBoost, and a novel hybrid CNN-LSTM model. The evaluation metrics are used to measure each algorithm's performance.

To assess the effectiveness of the various algorithms employed in the experiment, several statistical measures were utilized, including kappa statistics, precision, recall, f-measure, Matthew's correlation coefficient (MCC), receiver operating characteristic (ROC), and precision-recall (PRC). These evaluation matrices employ the terms true positive (TP), true negative (TN), false positive (FP), and false negative (FN). TP and TN refer to successfully recognized samples; meanwhile, FP and FN belong to instances that the models wrongly identified. The following provides a detailed elaboration of the mentioned evaluation metrics. These measures were used to compare the effectiveness of the algorithms and ensure the reliability and accuracy of the experiment's outcomes [3]. The formula for these methods is as follows:

$$Recall = \frac{TP}{TP + FN} \quad (6)$$

$$F1 - score = 2 \times \frac{TP + TN}{TP + TN + FP + FN} \quad (7)$$

$$Precision = \frac{TP}{TP + FP} \quad (8)$$

$$Sensitivity = \frac{TP}{TP + FN} \quad (9)$$

$$Specificity = \frac{TN}{TN + FP} \quad (10)$$

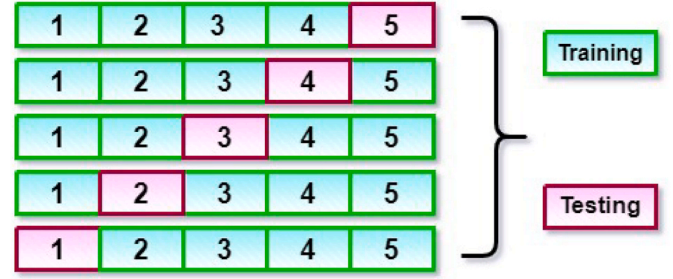


Fig. 7. Graphical representation of K-fold cross-validation.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (11)$$

5.3. Exploratory data analysis

In order to gain a better understanding of the dataset's characteristics, exploratory data analytics were performed. The results of these analyses are presented in the following section. Fig. 5 is a heatmap depicting the associated values and inter-feature correlations. All the colored cells show a correlation between the two characteristics and their associated values. The cell color indicates the strength of the connection, with negative values represented by a distinct color and zero indicating no correlation between variables.

In addition, Fig. 6 depicts the density distribution of a dataset of patients with and without illness. According to the dataset, the afflicted group consists of patients aged 50 to 60. The graph indicates that age is a significant risk factor for heart disease and that the likelihood of developing the condition grows.

5.4. Experimental results

In our study, many classification techniques, including GNB, SVM, DT, LR, CatBoost, AdaBoost, GBM, MLP, XGBoost, and CNN-LSTM, have been applied to our dataset and processed the features by applying feature engineering (FE) techniques and outliers have been found and removed. These classification algorithms were applied to the dataset using 5-fold cross-validation techniques; how it works is shown in Fig. 7.

Table 3 displays the classification method's sensitivity, specificity, and accuracy with and without FE. With FE, CatBoost, RF, and CNN-LSTM models give the highest accuracy. XGBoost, GBM, and CNN-LSTM parameters give the highest sensitivity, whereas CNN-LSTM, RF, and AdaBoost give the highest specificity. The results show that MLP, XGBoost, and CNN-LSTM achieved the highest accuracy after applying FE. CatBoost, GBM, and MLP achieved the highest sensitivity, while RF, AdaBoost, and CNN-LSTM achieved the highest specificity. However, our proposed model outperformed the other algorithms regarding overall performance.

Table 4 reveals that LR, SVM, and GNB under-performed compared to other algorithms when precision, recall, and f-measures were evaluated without FE. However, when FE was applied, DT, GBM, and CatBoost exhibited poor performance. On the other hand, RF, AdaBoost, and GNB showed exceptionally high performance without FE. With FE, CNN-LSTM, AdaBoost, and SVM demonstrated high performance.

The results of kappa statistics and MCC values for various classification systems, with and without FE, are presented in Table 5.

The findings indicate that our proposed model outperforms the other most effective and valuable algorithms. Table 6 displays each classification method's area under ROC and PRC values.

The area under ROC represents a common region between the TP and FP rates. In contrast, the area under PRC indicates a common area between precision and recall. Even though GNB, SVM, and LR get the closest results, AdaBoost, MLP, and CNN-LSTM perform better without

Table 3

Classification results of the different classification algorithms regarding Sensitivity, Specificity, and Accuracy.

Algorithm	Sensitivity		Specificity		Accuracy	
	Without FE	With FE	Without FE	With FE	Without FE	With FE
GNB	56.12%	69.80%	71.86%	75.79%	59.70%	72.16%
SVM	62.06%	69.48%	64.27%	76.79%	63.09%	72.27%
DT	71.07%	72.20%	74.22%	72.83%	72.55%	72.48%
LR	64.14%	71.84%	64.34%	75.33%	64.24%	73.32%
CatBoost	71.14%	73.48%	75.31%	73.64%	73.06%	73.55%
AdaBoost	69.01%	71.42%	78.47%	76.85%	72.84%	73.62%
GBM	72.01%	73.92%	73.69%	73.78%	72.83%	73.86%
RF	70.67%	72.43%	76.78%	75.96%	73.36%	73.93%
MLP	71.32%	74.21%	73.34%	72.90%	72.29%	74.06%
XGBOOST	72.02%	73.35%	73.48%	75.06%	72.74%	74.11%
Proposed CNN-LSTM	71.17%	72.04%	76.39%	77.11%	73.52%	74.15%

Table 4

Comparison based on precision, recall, and F-Measures.

Classifier Algorithm	Precision		Recall		F-Measure	
	Without FE	With FE	Without FE	With FE	Without FE	With FE
GNB	87.15%	81.59%	56.12%	69.80%	68.28%	75.23%
SVM	66.46%	82.91%	62.06%	69.48%	64.19%	75.60%
DT	75.63%	76.27%	71.07%	72.20%	73.28%	74.18%
LR	63.83%	79.80%	64.14%	71.84%	63.99%	75.61%
CatBoost	77.16%	76.62%	71.14%	73.48%	74.03%	75.02%
AdaBoost	82.46%	81.87%	69.01%	71.42%	75.14%	76.29%
GBM	74.26%	76.57%	72.01%	73.92%	73.12%	75.22%
Random Forest	79.45%	80.26%	70.67%	72.43%	74.80%	76.14%
MLP	74.14%	76.57%	71.32%	74.21%	72.70%	75.37%
XGBoost	73.93%	78.61%	72.02%	73.35%	72.97%	75.89%
Proposed CNN-LSTM	78.64%	81.82%	71.17%	72.04%	74.72%	76.62%

Table 5

Evaluation by kappa and MCC.

Classifier Algorithm	kappa		Mcc	
	Without FE	With FE	Without FE	With FE
GNB	19.61%	43.89%	23.46%	44.59%
SVM	26.21%	44.06%	26.27%	44.98%
Decision Tree	45.12%	45.12%	45.21%	44.85%
Logistic Regression	28.48%	46.35%	28.48%	46.66%
CatBoost	46.13%	46.94%	46.30%	46.99%
AdaBoost	45.73%	46.88%	46.62%	47.44%
GBM	45.66%	47.58%	45.69%	47.61%
RF	46.76%	47.58%	47.12%	47.88%
MLP	44.59%	48.00%	44.63%	48.02%
XGBoost	45.48%	48.01%	45.49%	48.14%
Proposed CNN-LSTM	47.07%	47.97%	47.33%	48.41%

Table 6

Value of area under ROC and PRC.

Classifier Algorithm	AUROC		AUPRC	
	Without FE	With FE	Without FE	With FE
GNB	0.5982	0.7160	0.6733	0.76254
SVM	0.6311	0.7060	0.6935	0.7346
DT	0.7256	0.7234	0.7653	0.7593
LR	0.6423	0.7213	0.7726	0.7592
CatBoost	0.7307	0.7342	0.7858	0.7801
AdaBoost	0.7288	0.7292	0.7766	0.7717
GBM	0.7283	0.7373	0.7886	0.7828
RF	0.7326	0.7370	0.7894	0.7804
MLP	0.7273	0.5273	0.7787	0.6081
XGBoost	0.7274	0.7383	0.7862	0.7829
Proposed CNN-LSTM	0.7344	0.7395	0.7873	0.7829

FE. Using FE, MLP, SVM, and GNB get the closest results, while GBM, XGBoost, and CNN-LSTM demonstrate superior performance.

Fig. 8a & b depicts the ROC curve of different classification algorithms with and without FE, constructed from the true and false positive rates. It is a graphical depiction of Table 6's area under ROC (AUROC). Fig. 9a and b illustrate the area under the precision-recall curve (AUPRC) for several classification methods, with and without FE, respectively. The AUPRC values represent the location and agree with the values reported in Table 6. These figures provide a visual representation of the AUPRC values. Table 7 displays the five most relevant characteristics based on correlation value and feature relevance.

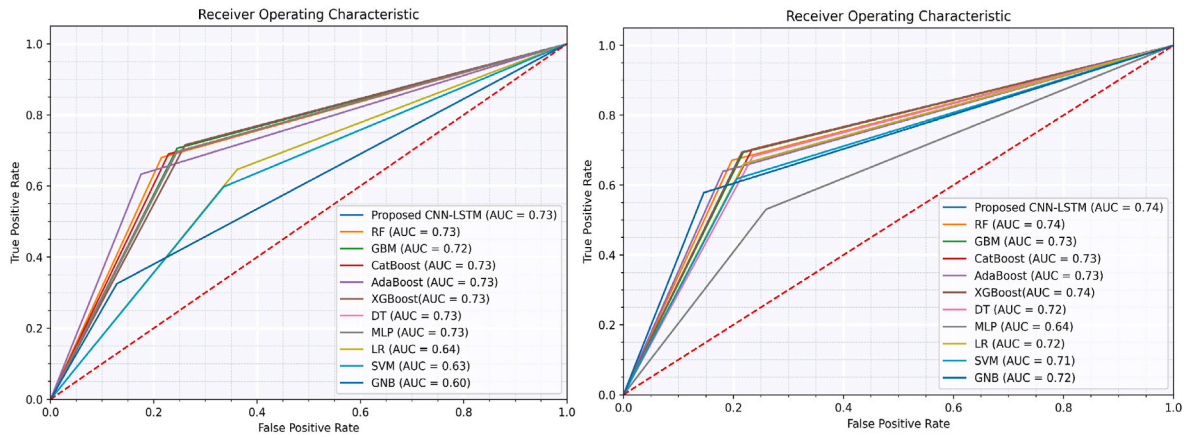
According to the table, high arterial pressure (ap_hi) is the significant trait or factor for identifying and predicting heart disease with and without FE for various classification algorithms.

In addition to age, the level of cholesterol in the blood (cholesterol), the patient is either underweight or overweight or normal (BMI), and the alcohol level in the patient's blood (alco) are other important predictors of heart disease.

According to the table, high arterial pressure (ap_hi) is the significant trait or factor for identifying and predicting heart disease with and without FE for various classification algorithms. In addition to age, the level of cholesterol in the blood (cholesterol), the patient is either underweight or overweight or normal (BMI), and the alcohol level in the patient's blood (alco) are other important predictors of heart disease.

Table 8 displays applicable classification algorithms' feature importance and coefficient scores, excluding SVM, MLP, and GNB, which do not produce such values. The importance and coefficients of each feature are presented in the table. Fig. 12a and b visually represent the important features of our proposed CNN-LSTM model as presented in Table 8. The figures illustrate the feature ranking based on their significance and coefficient scores, providing insight into the significant risk factors for CVD.

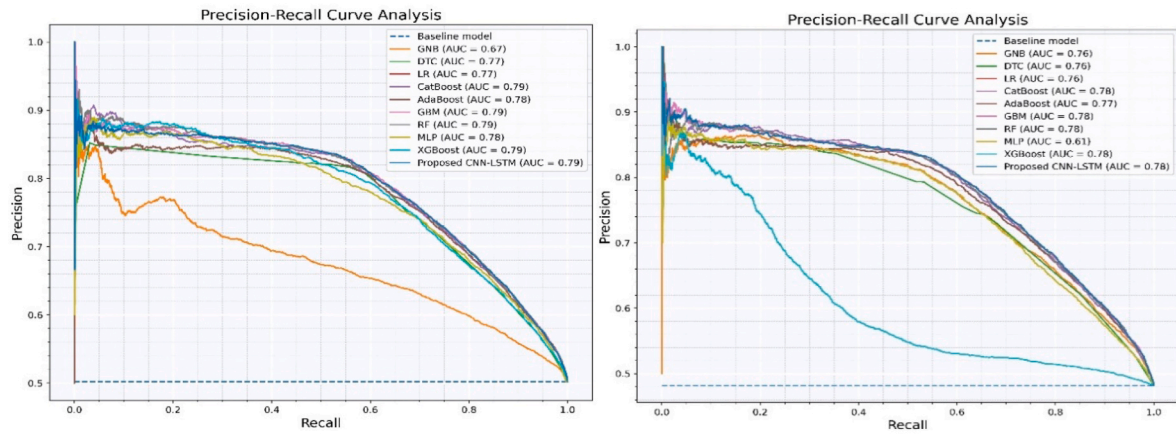
Without FE, the classifiers had an accuracy range of 59.70%–73.52%. However, when utilizing the same splitting with FE, the



(a) ROC curve analysis without FE

(b) ROC curve analysis with FE

Fig. 8. ROC curve analysis obtained from the proposed hybrid CNN-LSTM model.



(a) PRC curve analysis without FE

(b) PRC curve analysis with FE

Fig. 9. PRC curve analysis obtained from the proposed hybrid CNN-LSTM model.

Table 7

Top five features for heart disease according to applied algorithms.

Feature Ranking		1st	2nd	3rd	4th	5th
DT	With FE	blood_diff	ap_hi	ap_lo	cholesterol	BMI
	Without FE	ap_hi	age	cholesterol	alco	gluc
CatBoost	With FE	ap_hi	cholesterol	ap_lo	active	BMI
	Without FE	ap_hi	age	cholesterol	ap_lo	weight
AdaBoost	With FE	ap_hi	cholesterol	BMI	active	weight
	Without FE	age	ap_hi	weight	cholesterol	ap_lo
GBM	With FE	ap_hi	cholesterol	BMI	ap_lo	weight
	Without FE	ap_hi	age	cholesterol	weight	ap_lo
RF	With FE	ap_hi	ap_lo	blood_diff	cholesterol	BMI
	Without FE	ap_hi	ap_lo	age	cholesterol	weight
XGBoost	With FE	ap_hi	cholesterol	ap_lo	active	smoke
	Without FE	ap_hi	age	cholesterol	ap_lo	active
Proposed CNN-LSTM	With FE	age	ap_hi	hypertense	ap_lo	blood_diff
	Without FE	ap_hi	ap_lo	age	cholesterol	weight

classifiers' accuracy ranged from 72.16% to 74.15%. Additionally, we evaluated our proposed CNN- LSTM classifier alongside all the ML classifiers used in our experiment. Table 9 presents the results obtained with and without FE.

6. Discussions

The results presented in Table 9 show that our proposed CNN-LSTM model outperformed all other classification algorithms regarding accuracy, precision, recall, and F1-score, with and without FE. The accuracy of the proposed model was 73.89% without FE, which increased to 74.95% with FE. Moreover, the proposed model's precision, recall, and

Table 8

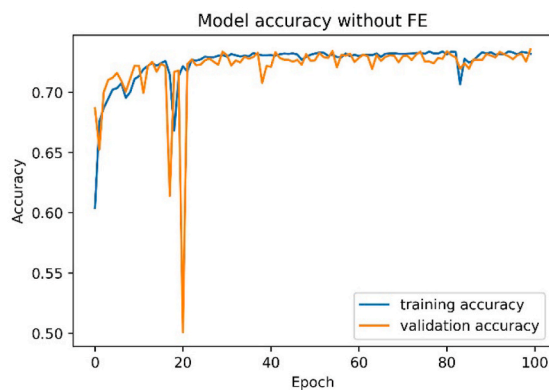
Feature importance and coefficient scores of different applied algorithms.

Algorithm		DT	CatBoost	AdaBoost	GBM	RF	XGBoost	Proposed CNN-LSTM
age	Without FE	0.1031	24.0725	0.22	0.1233	0.1253	0.1472	0.0363
	With FE	0.0	0.0	0.0	0.0	0.0	0.0	0.0
gender	Without FE	0.0013	0.0	0.0	0.0010	0.0050	0.0184	0.0184
	With FE	0.0	0.5600	0.0	0.0003	0.0042	0.0047	0.0047
height	Without FE	0.0010	0.2378	0.01	0.0050	0.0307	0.0046	0.0046
	With FE	0.0	0.3025	0.005	0.0026	0.0225	0.0033	0.0033
weight	Without FE	0.0042	5.3016	0.15	0.0230	0.0307	0.0288	0.0288
	With FE	0.0096	1.4251	0.07	0.0089	0.0448	0.0117	0.0117
ap_hi	Without FE	0.7926	42.7182	0.23	0.7345	0.4731	0.5227	0.5226
	With FE	0.2856	65.4973	0.23	0.8060	0.3503	0.6932	0.6932
ap_lo	Without FE	0.0034	7.1226	0.075	0.0202	0.1785	0.0681	0.0681
	With FE	0.5394	3.2506	0.065	0.0171	0.1763	0.0411	0.0411
cholesterol	Without FE	0.0798	17.7410	0.135	0.0745	0.0930	0.1120	0.1193
	With FE	0.0746	19.7296	0.165	0.1008	0.0945	0.1295	0.1295
gluc	Without FE	0.0072	0.0	0.05	0.0061	0.0142	0.0119	0.0119
	With FE	0.0	2.2176	0.065	0.0073	0.0144	0.0166	0.0166
smoke	Without FE	0.0000	1.0115	0.03	0.0023	0.0051	0.0250	0.0250
	With FE	0.0	0.8309	0.045	0.0037	0.0049	0.0187	0.0187
alco	Without FE	0.0080	0.0	0.035	0.0025	0.0049	0.0172	0.0172
	With FE	0.0	0.6074	0.055	0.0027	0.0041	0.0139	0.0139
active	Without FE	0.0064	1.7944	0.065	0.0073	0.0089	0.0363	0.0363
	With FE	0.0008	2.8462	0.075	0.0085	0.0084	0.0342	0.0342

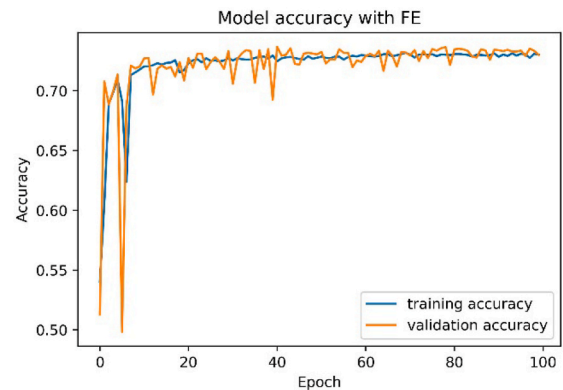
Table 9

Results comparison of different ML algorithms against our proposed CNN-LSTM classifier using a 5-fold cross-validation technique.

Algorithms	1st fold C (%)		2nd fold C (%)		3rd fold C (%)		4th fold C (%)		5th fold C (%)		CV Means (%)		CV STD (%)	
	Without FE	With FE	Without FE	With FE	Without FE	With FE	Without FE	With FE	Without FE	With FE	Without FE	With FE	Without FE	With FE
GNB	0.596	0.715	0.595	0.716	0.587	0.713	0.593	0.712	0.594	0.711	0.593	0.713	0.003	0.0019
SVM	0.677	0.483	0.689	0.480	0.679	0.489	0.689	0.489	0.685	0.491	0.684	0.486	0.0051	0.0043
DT	0.727	0.727	0.732	0.732	0.733	0.733	0.727	0.727	0.723	0.723	0.729	0.728	0.0036	0.0037
LR	0.714	0.719	0.725	0.724	0.718	0.720	0.729	0.723	0.718	0.7111	0.721	0.720	0.0055	0.0044
CatBoost	0.731	0.727	0.739	0.735	0.733	0.734	0.732	0.732	0.729	0.727	0.733	0.731	0.0033	0.0031
AdaBoost	0.729	0.508	0.731	0.732	0.729	0.729	0.726	0.727	0.725	0.491	0.728	0.638	0.0024	0.1127
GBM	0.732	0.509	0.743	0.739	0.735	0.734	0.737	0.734	0.733	0.648	0.736	0.672	0.0040	0.0885
RF	0.527	0.676	0.738	0.736	0.736	0.731	0.73	0.731	0.716	0.730	0.690	0.721	0.082	0.0222
MLP	0.728	0.518	0.742	0.492	0.732	0.491	0.734	0.659	0.729	0.509	0.733	0.534	0.0048	0.0635
XGBoost	0.731	0.738	0.743	0.740	0.734	0.732	0.737	0.735	0.734	0.732	0.736	0.734	0.0039	0.0041
Proposed CNN-LSTM	0.736	0.742	0.743	0.746	0.736	0.739	0.732	0.738	0.734	0.744	0.746	0.749	0.0042	0.0040



(a) Model accuracy curve without FE.



(b) Model accuracy curve with FE.

Fig. 10. Accuracy curve analysis of our proposed hybrid CNN-LSTM model.

F1-score were 74.02%, 73.72%, and 73.84%, respectively, without FE, which improved to 75.09%, 75.22%, and 75.15%, respectively, with FE. These results indicate that the proposed CNN-LSTM model more effectively predicts CVD risk when the FE gets associated with it. The

study conducted an evaluation of the proposed CNN-LSTM model, comparing its performance with and without FE. Figs. 10a and 11a illustrate the outcomes of the model without FE, indicating a test accuracy of 73.52% and the convergence of training and validation loss

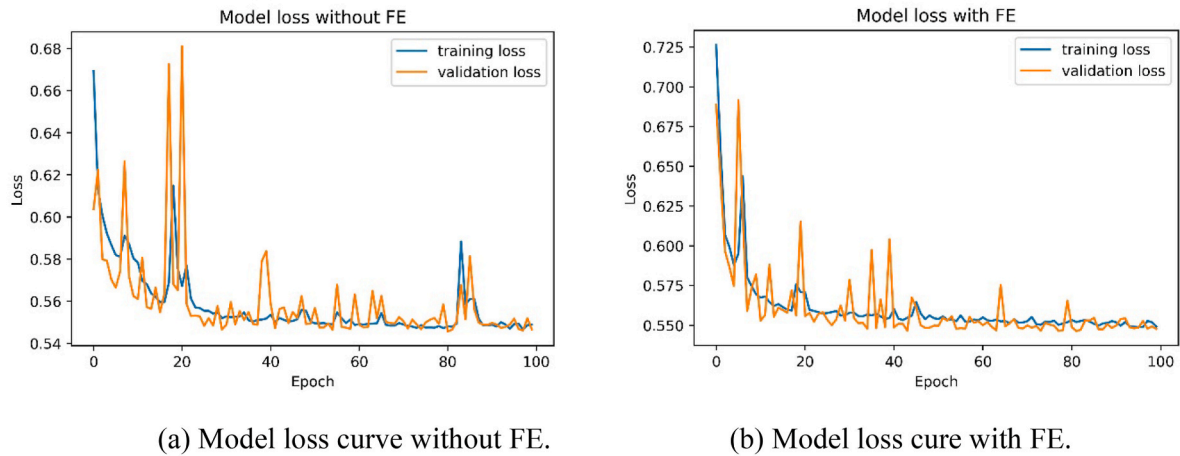


Fig. 11. Model loss curve analysis of our proposed hybrid CNN-LSTM model.

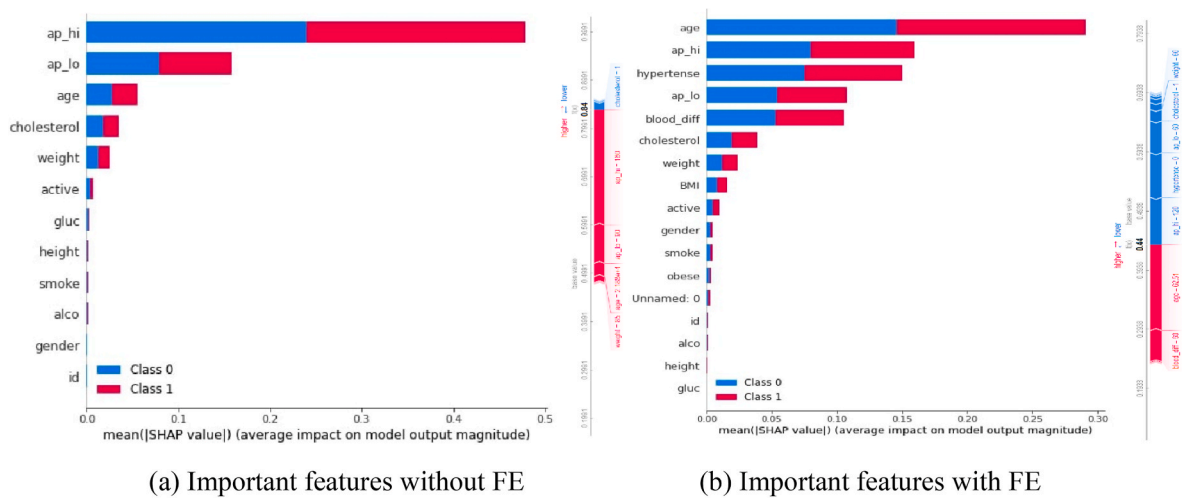


Fig. 12. Important features according to SHAP using our proposed CNN-LSTM model.

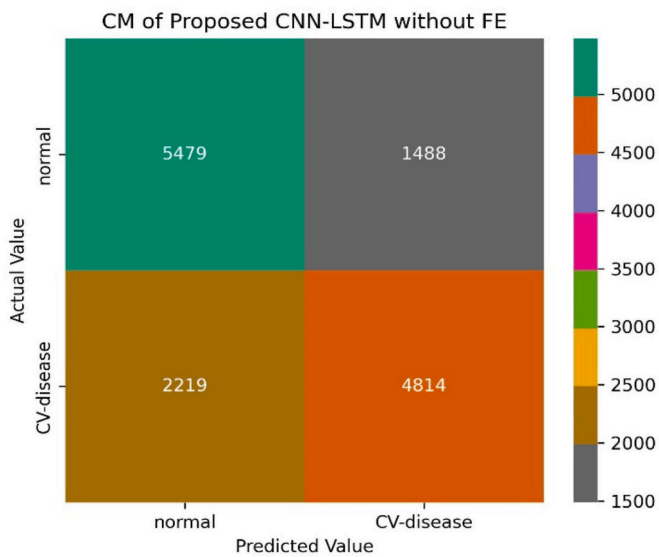


Fig. 13. The confusion matrix of our proposed CNN-LSTM without FE takes 80% as training and 20% as testing data.

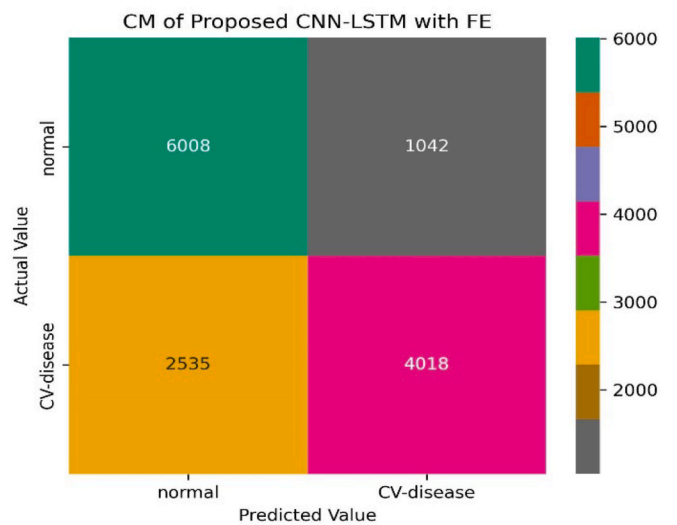


Fig. 14. The confusion matrix of our proposed CNN-LSTM with FE takes 80% as training and 20% as testing data.

Cardiovascular Disease Identification using CNN-LSTM

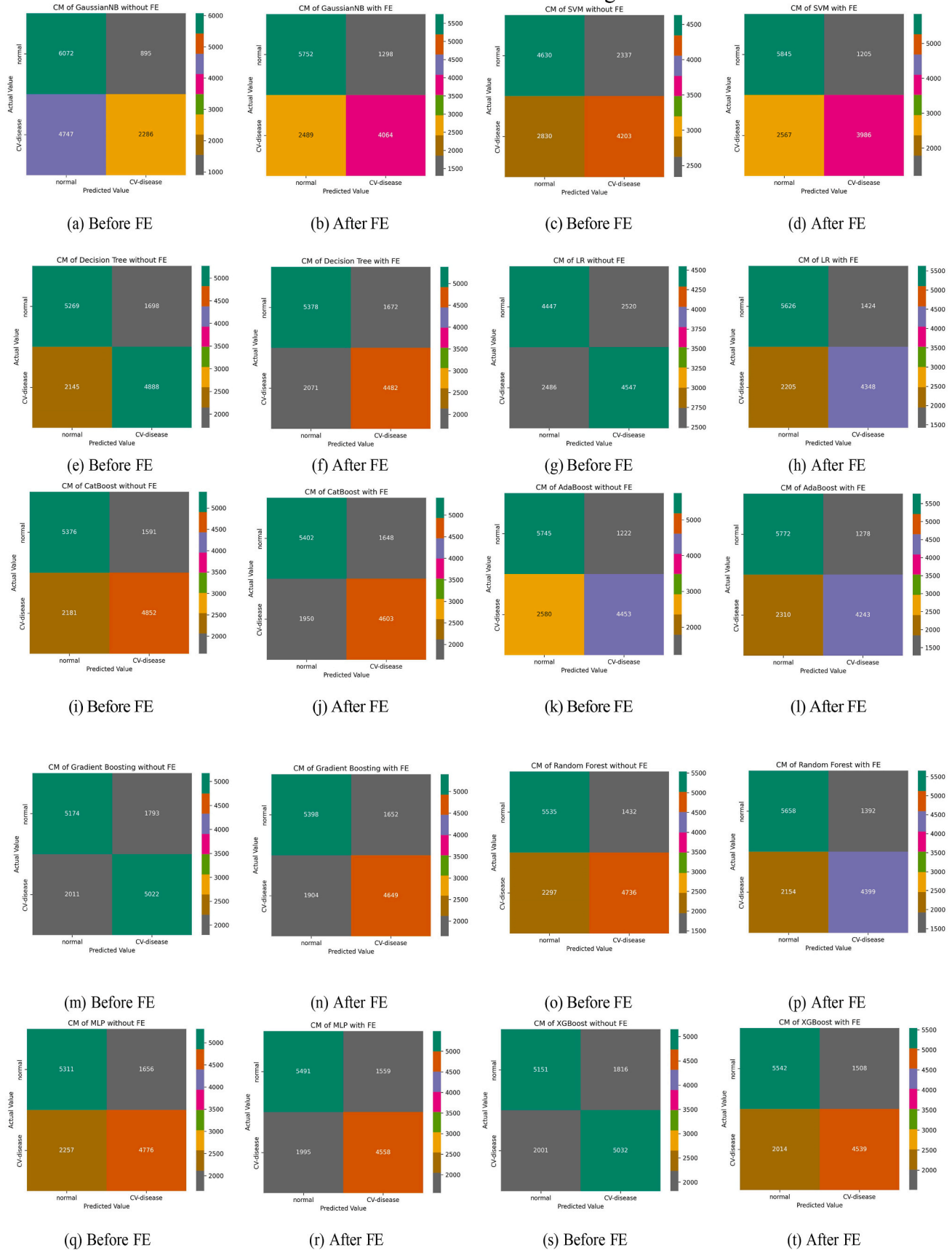


Fig. 15. The confusion matrix of all the experimented classifiers on Dataset before feature engineering (FE) and after feature engineering (FE), respectively.

Table 10

Comparison between the state-of-the-art approaches and our proposed CNN-LSTM model on the same dataset.

Study	Precision	Recall	F- Measure	Kappa	Mcc	AUROC	Sensitivity	Specificity	Accuracy
[1]	×	×	72%	×	×	×	×	×	72.70%
Our study	81.82%	72.04%	76.62%	47.97%	48.41%	73.95%	72.04%	77.11%	74.15%

after the 55th epoch. Subsequently, Figs. 10b and 11b present the network's performance on training and validation data with FE.

These figures provide a comprehensive overview of accuracy and loss for each epoch. With FE, the CNN-LSTM model exhibited convergence at the 82nd epoch, achieving the highest validation score. The model demonstrated a classification performance of 74.15% on unseen test data not encountered during training. These results indicate that FE enhances the CNN-LSTM model's performance, improving accuracy and convergence. The findings emphasize incorporating FE techniques in the proposed model for enhanced predictive capabilities. The significance of individual features in a model can be determined by analyzing their absolute Shapley values. The absolute Shapley values for each feature are averaged across the dataset to obtain a global understanding of feature importance. Afterwards, the features are sorted in descending order of importance, and a plot is generated to visualize the results. In this context, the proposed CNN-LSTM model was utilized to predict CVD. Fig. 12a depicts the Shapley values for important features without employing FE techniques. The plot indicates that the most significant feature is *ap_hi*. Conversely, Fig. 12b shows the important features after applying FE techniques, and the plot reveals that age is the most significant feature, followed by *ap_hi*.

In our study, we utilized the confusion matrix to compare the performance of the proposed CNN-LSTM model with and without FE. Fig. 13 displays the confusion matrix obtained from the CNN-LSTM model without FE.

The confusion matrix shows that the model correctly predicted 5479 instances while the TN was 4814. Fig. 14 displays the confusion matrix obtained from the CNN-LSTM model with FE, showing that the model correctly predicted 6008 instances while the TN was 4018. Overall, adding FE led to an increase in the number of accurate predictions, indicating that the model could better identify cases of heart disease.

Fig. 15 displays the dataset's confusion matrix. By dividing the dataset into training and testing halves, 20% of the data and 80% of the data were used to assess the performance of various classifiers.

The confusion matrix provides a valuable visual representation of the model's performance and can be used to guide further improvements to the model. Furthermore, it is worth noting that some classification algorithms, such as RF, AdaBoost, and XGBoost, achieved high accuracy and F1-score with FE. However, their performance decreased without FE, indicating that FE is crucial for these algorithms' performance.

On the other hand, MLP and CatBoost achieved high accuracy and F1-score with and without FE, indicating their robustness to FE. Our proposed CNN-LSTM model performs better in predicting cardiovascular disease (CVD) risk than other classification algorithms. One of the recent study [1] on the same dataset achieved an accuracy of 72.7%, while our model demonstrates an exceptional accuracy of 74.15% in diagnosing early-stage cardiac conditions. This represents a significant improvement of 1.45% over the previous state-of-the-art approaches.

Comparison between the state-of-the-art approaches and our proposed CNN-LSTM model on the same dataset, are shown in Table 10. However, in this study, the author did not employ explainable AI and feature engineering techniques to visualize the specific features responsible for the disease. We utilized these, which underscores the uniqueness of our study.

However, other algorithms such as RF, AdaBoost, XG-Boost, MLP, and CatBoost also achieved high accuracy and F1-score with FE. Therefore, FE is an essential step in improving the performance of these algorithms. Moreover, our study addressed these research gaps and limitations by utilizing a hybrid CNN-LSTM model, incorporating a

diverse dataset, and employing XAI and feature engineering techniques. These contributions enhance the development of effective and efficient ML models for CVD prediction, providing valuable insights for future research and clinical applications.

However, some limitations of this study should be acknowledged. First, the proposed model was evaluated on a single dataset, and its performance should be validated on other datasets. Second, the model was trained using a supervised learning approach, which requires a large amount of labeled data. Obtaining labeled data can be challenging, and alternative approaches, such as semi-supervised or unsupervised learning, may be necessary.

7. Conclusions

In conclusion, our study makes significant strides in the accurate prediction of heart disease using data mining and machine learning techniques. Through the evaluation of various ML algorithms, we have successfully demonstrated that the CNN-LSTM hybrid model emerges as the most effective algorithm, achieving a commendable accuracy of 74.15%. This finding highlights the potential of advanced ML models in the realm of cardiovascular disease (CVD) prediction, providing a valuable tool for early intervention and prevention. Moreover, we employed the SHAP technique to interpret the importance of features in the models, offering crucial insights into the underlying mechanisms of disease prediction. By identifying key features such as age, systolic blood pressure, and cholesterol levels as essential contributors to heart disease prediction, we have deepened our understanding of the factors influencing CVD development. This valuable information can potentially aid clinicians in devising personalized treatment plans and risk management strategies for patients, ultimately leading to improved clinical outcomes. Our study successfully addresses several research gaps and limitations by leveraging a hybrid CNN-LSTM model, utilizing a diverse dataset, and incorporating feature engineering techniques. These contributions enhance the effectiveness and efficiency of ML models for CVD prediction, positioning them as promising tools in the domain of cardiovascular health management. However, we acknowledge that further research is warranted to validate our findings on larger and more diverse datasets. Continued exploration of ML techniques and their clinical applications in predicting and preventing heart disease will undoubtedly strengthen the field's knowledge base and enable better-informed decision-making for medical practitioners.

CRediT authorship contribution statement

Md. Maruf Hossain: Conceptualization, Methodology, Software, Validation, Writing - Original Draft, **Md Shahin Ali:** Conceptualization, Methodology, Software, Validation, Writing - reviewing & editing. **Md. Mahfuz Ahmed:** Data curation, Writing, Validation. **Md. Rakibul Hasan Rakib:** Data curation, Writing, Validation. **Moutushi Akter Kona:** Data curation, Writing, Validation. **Sadia Afrin:** Data curation, Writing, Validation. **Md Khairul Islam:** Methodology, Formal analysis, Writing - Review & Editing and Supervising. **Md Manjurul Ahsan:** Investigation, Writing - Review & Editing. **Sheikh Md. Razibul Hasan Raj:** Data curation, Writing, Validation. **Md Habibur Rahman:** Formal analysis, Writing - Review & Editing and Supervising.

Data availability

The data used to support the findings of this study are available from

the corresponding author upon request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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